

# AI ASSISTED CODING LAB – 1

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## Lab Assignment -1: Comparing Three Ai Codes

**Prompt:** Write a python code of prime numbers from 1 to 20

### Github Copilot:

```
primes = []
for num in range(2, 21):
    is_prime = True
    for i in range(2, int(num ** 0.5) + 1):
        if num % i == 0:
            is_prime = False
            break
    if is_prime:
        primes.append(num)

print("Prime numbers from 1 to 20:")
print(primes)
```

### #output

Prime numbers from 1 to 20:

[2, 3, 5, 7, 11, 13, 17, 19]

## Goggle Collab

```
#code def is_prime(num):    if num
<= 1:        return False    for i in
range(2, int(num**0.5) + 1):        if
num % i == 0:            return False
return True
```

```
prime_numbers = [num for num in range(1, 21) if
is_prime(num)] print(prime_numbers)
```

### **#output**

```
[2, 3, 5, 7, 11, 13, 17, 19]
```

## Claude

### **#code**

```
def is_prime(n):
    if n < 2:
        return False    for i in
range(2, int(n**0.5) + 1):        if n
```

```
% i == 0:      return False  
  
return True
```

```
primes = [n for n in range(1, 21) if is_prime(n)]  
print(primes)
```

### **#Output**

```
[2, 3, 5, 7, 11, 13, 17, 19]
```

### **Conclusion:**

The three AI-generated Python codes successfully identify and display the prime numbers from 1 to 20. GitHub Copilot, Google Colab, and Claude used slightly different coding approaches, but all produced the same correct output: [2, 3, 5, 7, 11, 13, 17, 19]. This comparison shows that AI tools can generate accurate and efficient Python programs, while also demonstrating that different AI tools may use different programming styles to solve the same problem.